

The Study of Coherent Signaling Across Systems

Linguistics is the study of how systems under energetic constraint and irreducible uncertainty maintain coherent signaling. Coherence is achieved through minimum-energy, systematically biased, lossy representations that are just good enough to support viable action and coordination. These representations exploit and constrain intrinsic variability rather than eliminating it. This essay is part of a series; the companion piece on randomness as structured potential should be read first.

1. Coherence, Syntax, and the Thermostat

The idea of communication is intuitive: one system affects another. The idea of language is less so. And the role of environmental feedback — the context that shapes meaning — is rarely acknowledged at all.

A mechanical thermostat communicates with a heating system through a binary signal: on or off. The thermostat registers the environment and relays that reading; the system responds, which shapes the next signal. The cycle forms a feedback loop. This call-and-response structure appears at every level of signaling — in conversation, where a question invites an answer; in music, where a phrase is echoed or resolved. It is not the signal alone that matters. It is the response within context. If any part of the loop breaks, the signal becomes noise.

Syntax, in its most general sense, refers to locally accumulated regularities in signal timing, density, and sequencing that allow coordination without requiring a central shared model. Even the thermostat has syntax of this kind: heat and cool cannot activate simultaneously; the system expects a certain structure, and coherence collapses without it. This is not the combinatorial, generative syntax of human language — that is a late specialisation, not the baseline. At every other level, syntax is statistical and historical: regularities shaped by selection and drift, not abstract rules imposed from above. Syntax enables coordination by constraining variability just enough to make useful action possible. It is scaffolding, not blueprint.

In human speech, word order disambiguates meaning. In music, syntax is rhythm, phrase structure, tension, and release. In bacterial quorum sensing — covered in the companion randomness essay — the timing and density of chemical signals follow thresholds: too early or too late, and the collective behavior fails to cohere. The medium changes; the underlying logic does not.

This points toward a working definition: linguistics is the study of coherent signaling across systems — achieved through minimum-energy, lossy representations that exploit and constrain intrinsic variability, enabled by locally accumulated regularities in signal structure, shaped by context, and sustained through feedback.

2. Dialects in the Wild: Frogs and Birds

If the thermostat illustrates the minimum conditions for signaling, frogs and birds illustrate how signaling systems evolve — and why variability is not a problem to be solved but a resource to be managed. Their calls are not mere noise; they are dialects shaped by natural selection and random drift, adapting continuously to changing environments.

In species such as the Pacific tree frog, call variations emerge across habitats: higher pitches in noisy urban settings, lower tones in quieter forests. Females prefer calls that stand out against background noise, so clearer signals yield better mates and stronger offspring (Gerhardt & Huber, 2002). Isolated populations develop unique call patterns over generations even without clear environmental pressure — drift producing accent where selection produces adaptation. These drifting dialects are not failures of coherence; they are partially decoupled explorations whose local coherence is cheap to maintain and whose divergence preserves usable variability.

across the population. The dialect is a shock absorber as much as a signal.

Songbirds amplify this story. White-crowned sparrows adjust their melodies to local conditions, and dialects diverge in isolated groups. Crucially, dialects reflect both genetic isolation and cultural transmission — passed from tutor to fledgling across generations (Nelson & Soha, 2004). What ties frogs and birds to bacteria is the same underlying logic: coherence that fits the context, achieved cheaply, while preserving the variability that makes future adaptation possible. Stereotyped, efficient signaling is the constrained special case. Variability is the default.

3. Voices: The Efficiency of Auditory Signaling

Auditory communication is remarkably widespread across species, and for good reason. Sound travels approximately 1,500 m/s in water — outpacing visual cues substantially in murky environments. Early life forms likely used hair-like structures to sense fluid vibrations, establishing a foundational pattern of mechanosensation that eventually produced the vertebrate ear.

Voices, however, are not an obvious consequence of hearing. A fish grunt or bird chirp risks drawing predators. Vocal apparatus and the neural circuits supporting it are metabolically expensive. The payoff must be substantial, and it is: auditory signals abstract complex needs — mating, territory, alarm — into fast, reliable signals that cut through environments where visual communication fails. The cost is real; so is the return.

The structural parallel to syntax is direct. Fish sounds are not random: toadfish hums follow rhythmic patterns; damselfish chirps carry territorial meaning; herring produce sounds that enhance group cohesion. These signals are structured within biological constraints — anatomical, evolutionary — exactly as bacterial chemical signals are structured within chemical ones. The regularities are local and historically accumulated, not centrally designed. The medium changes; the architecture does not.

4. Cosmic Logic: Randomness as the Precondition for Meaning

Before grammar, before thought — before even life itself — there was potential. The cosmos is not logical in the way we once hoped. It is recursive, probabilistic, and emergent. Order and meaning arise not from being imposed but from filtering noise through constraint, feedback, and time. Language, like life, lives here — in the middle, between zero potential and infinite potential. Shaped by selection, not by design.

Physicists now understand that without quantum fluctuations in the early universe — random deviations from perfect uniformity — nothing would exist. These fluctuations, amplified during cosmic inflation, became the seeds of galaxies. Randomness did not distort perfection. It made reality possible. The companion essay on randomness develops this principle in detail. What matters here is its implication for signaling: coherence is not extracted from order. It is constructed from structured uncertainty, temporarily and at energetic cost, within a sea of variability that never goes away. Variability is the prior state; coherence is what selection and learning impose on it when the environment rewards them consistently enough.

Our minds evolved in a world of rocks and predators, not wave functions and nonlocality. We are wired for cause and effect, not entangled states. Abstract reasoning is a late evolutionary invention — a tool for navigating a world built on patterns we can never fully feel. The next section examines the specific biological event that made that tool possible.

5. FOXP2: From External Search to Internal Recursion

Charles Darwin understood systems before mechanisms. He had no knowledge of DNA or transcription factors, yet his theory of evolution still stands. That same humility is warranted when discussing the origins of language: we can observe the pattern without fully cracking the mechanism.

The FOXP2 gene is found across vertebrates. What distinguishes humans is just two amino acid substitutions relative to chimpanzees — minor differences with outsized effects. They reshape neural circuitry, particularly the cortico-basal ganglia loops associated with motor control and sequential learning. In mice engineered to carry the human variant, these changes produce longer dendrites, greater synaptic plasticity, altered dopamine signaling, enhanced motor sequencing — and reduced exploratory behavior.

That last point is the key. The humanized mice were better at solving mazes but explored less. In a lab, food is predictable. In the wild, it is not. What the humanized FOXP2 reveals is a shift in where the search happens. Before: in the world, by trial and error — external variability doing the work. After: in the brain, by recursion and simulation — internal modeling replacing physical exploration. The correspondence cost rises sharply. A richer internal model must be maintained and updated; it can drift from reality in ways that direct engagement cannot. That is where complex language begins — as internal abstraction, a recursive model of reality, foraging inside the skull. It is powerful. It is also expensive, and it narrows certain kinds of contact with the world.

In humans, FOXP2 mutations are linked to childhood apraxia of speech: a breakdown in the motor planning required for syllable sequencing. Control over lower-level motor functions is the prerequisite for turning vocalizations into symbols. The shift from external search to internal recursion is the biological root of the map/territory problem. Our words help us predict, categorize, and plan — but they can also blind us to what cannot be named, and to the irreducible processes that the model simplifies away. They are tools, not truths.

6. Wittgenstein's World

“The limits of my language mean the limits of my world.” Wittgenstein inhabited a philosophical world of language, logic, and mathematics — often artificially isolated from physical reality. The preceding sections have attempted to reconnect that world to its biological origins.

Mathematics is the most precise language we have. Its definitions are strict and unchanging — a sealed laboratory of symbols, untouched by noise. But as the randomness essay establishes, reality is not Newton's clockwork. It is quantum, chaotic, computationally irreducible. Accuracy sometimes demands that we sacrifice precision in order to move closer to the target. The felt limits of language — the sense that some experiences or processes resist naming — are not failures of vocabulary. They are the phenomenological consequence of the FOXP2 trade-off: a brain that models the world internally will always encounter processes that outrun the model. A theory of language tested only against itself mistakes the map for the terrain.

7. Einstein vs. Weinberg: Situated Intelligence and Symbolic Reduction

What we strive for in language is not precision but accuracy — and accuracy in a messy world requires situated intelligence. Knowing a genetic mechanism means little until it is placed inside a behavioral system. Knowing a word means little until it is placed inside a context. Meaning arises from how symbols are situated in reality, not from the symbols alone.

Einstein represents situated intelligence. His thought experiment of riding a beam of light is intuitive abstraction embedded in physical reality — perhaps the last great act of that kind before physics passed fully into the realm of pure mathematics. Einstein still paid the correspondence costs of embodiment: his intuitions had to survive experimental friction, contact with a world that pushes back. Relativity holds — within its environmental bounds. It was not wrong. It was incomplete. Like language, it works until the chaos pushes back hard enough.

Weinberg represents symbolic intelligence at its most powerful — a command of mathematics that risks losing contact with those correspondence costs entirely. Weinberg-style symbolic reduction tries to minimize the costs of embodied engagement through precision, but risks losing accuracy when the map drifts too far from the irreducible processes it describes. Symbolic systems help us act, but they also constrain what we can see. Precision needs friction.

8. Conclusion: Maps and Territory

Language is abstract. It is symbolic compression. And it can only be grounded in experience. The idea of a thing is not the thing itself.

From bacterial quorum sensing to bird-song dialects, from FOXP2 mutations to Einstein's thought experiments, the same recursive pattern appears at every level of complexity: systems filtering noise through constraint, exploiting variability rather than eliminating it, adapting signals to context, building models of a world they can never fully grasp. Randomness is not the enemy of meaning — it is its precondition. Coherence is not imposed; it emerges, temporarily, at cost, from a variability that is always the deeper truth.

In the Critique of Pure Reason, Kant argued that space and time are not properties of the world itself but formal constraints of our perception. We do not see things-in-themselves — we perceive appearances filtered through the structure of our cognition. Just as space and time filter perception, language — whether FOXP2 recursion or $E = mc^2$ — filters reality into ideas. The correspondence cost is always there, whether we account for it or not.

Fish communicate in pulses. Humans say ‘danger.’ Einstein builds spacetime curves. Weinberg writes wave functions. All of them, as Kant warned, miss the thing-in-itself. The best we can do is map what we are able to perceive — remain honest about the cost of the map — and remember that the terrain was variability all along. This is just one of a series of notes. Refer to the master index for an overview.

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